RoKGDS: A Robust Knowledge Grounded Dialog System

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Abstract. In this paper, we propose a pre-training based **R**obust **K**now-ledge **G**rounded **D**ialog **S**ystem (**RoKGDS**) to enhance the performance of the model in unknown scenarios, which is easily generalized to various knowledge grounded dialog tasks, such as persona dialog, knowledge dialog, recommendation dialog. We use a bucket encoder to efficiently extract all kinds of knowledge information (e.g. profile, knowledge graph, and dialog goal). To improve the robustness of the model, we develop a hybrid decoder with a hybrid attention and a copy mechanism. The hybrid attention is an adaptation scheme to apply the pre-trained language model to our model and the copy mechanism is a gate mechanism to control generating a word from generic vocabulary or the input knowledge. Experiments show that our model is more robust than the other baseline models. Furthermore, we use visualization to explain the effectiveness of the hybrid attention compared to other two adaptation schemes. In the 2021 Language and Intelligence Challenge: Multi-Skill Dialog task, our best model ranked 3rd in the automatic evaluation stage and 5th in the human evaluation stage.

Keywords: Knowledge grounded dialog system · Transfer learning

1 Introduction

Open-domain dialog system has recently attracted much attention [6], thanks to the success of end-to-end deep models trained with large-scale dialog data. However, these data-driven models face the problem of generating meaningless or repeated responses [8,9]. To improve the informativeness of generated responses, previous works grounded the conversation with topic [18], persona [19], knowledge graph [4,17] or dialog goal [10]. These knowledge grounded dialog tasks aim

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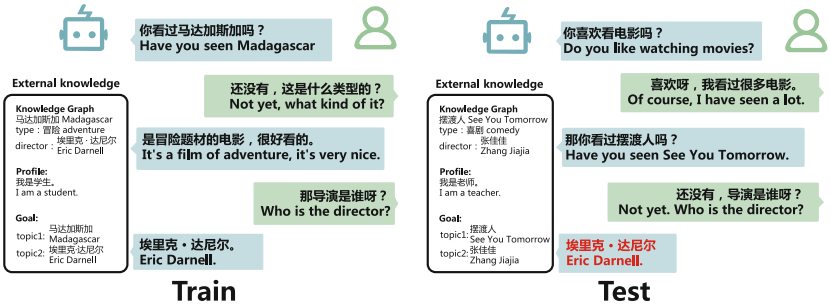

Fig. 1. A sample of knowledge grounded dialog. The external knowledge includes user profile, knowledge graph, and dialog goal. In the test scenario, the model cannot effectively use the knowledge that has never been exposed before, which leads to the failure of the dialog.

to use structured or unstructured knowledge information to accomplish persona consistent dialog, knowledge-accurate dialog, or purposeful dialog.

Although there are many dialog datasets with additional information to address knowledge grounded dialog tasks, such as *PersonaChat* [19], *DuConv* [4] and *DuRecDial* [10], these datasets are relatively small compared to open-domain dialog datasets without external knowledge, which leads the model to overfit the training data and not generalize well to the test data. Due to the small-scale training data, the model will mistakenly learn the mapping of the context and the knowledge that often appears in the responses, rather than the mapping of the context and the corresponding knowledge in the knowledge graph. For example in Fig. 1, the model tends to generate the response with the greatest likelihood in training data rather than the correct knowledge in the test scenario.

Recently, the pre-trained language models have achieved great success in the dialog generation task, which has significantly improved the model's robustness. However, most Chinese pre-trained language models (e.g. ERNIE [14], GPT-2 [12], BERT [3], RoBERTa [2], etc.) are based on either decoder-only or encoder-only Transformer [15]. Budzianowski and Vulić [1] use a common encoding method that concatenates all the additional information together (called *naive encoding*) as the input of the Transformer to apply these models to dialog task with additional information. However, it is time-consuming and space-consuming. It is difficult to apply these pre-trained language models to the dialog generation tasks with a large amount of knowledge.

To address the problems above, we propose a general robust knowledge grounded dialog system (RoKGDS) based on RoBERTa. Since there is a lot of input knowledge, such as user profile, knowledge graph, or dialog goal, RoKGDS first uses a bucket encoder to efficiently encoding the knowledge information. Then a hybrid decoder is developed to extract useful knowledge information conditioned on the dialog context through a hybrid attention. Finally, RoKGDS use a copy mechanism to copy correct knowledge from the input knowledge

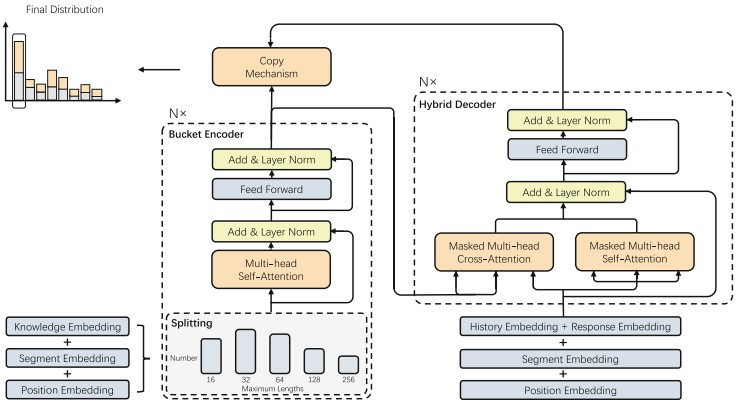

Fig. 2. The architecture of the RoKGDS

and avoid generating unrelated knowledge. Both automatic and manual evaluations on *DuConv*, *DuRecDial*, *CPC* all show that RoKGDS is more robust than other baseline models. Moreover, we visualize the attention score between context and knowledge to study the behavior of the hybrid attention compared to the other two baseline adaptation schemes in response generation, indicating that RoKGDS can effectively incorporate the given knowledge and context in the test scenario.

2 Related Work

Recently, the pre-trained language models such as BERT, GPT-2, ERNIE, etc. have led to state-of-the-art results in many tasks. In the dialog generation task, the pre-trained language models improve the model's ability to understanding and generating natural language. Wolf et al. [16] proposed to combine transfer learning with multi-task learning to improve discriminative language understanding. Budzianowski and Vulić [1] joined the belief state, context, and database state together as the GPT-2's input to accomplish task-oriented dialog. Golovanov et al. [5] studied using different finetune strategies to apply GPT-2 to improve the quality of the generated responses on persona dialog task.

Compared to the previous works, we propose a general model to improve the model's robustness in knowledge grounded dialog, which is easy to be generalized to other similar tasks. Additionally, our work evaluates the adaptation schemes of the pre-trained model on three tasks with additional knowledge – knowledge dialog, recommendation dialog, and persona dialog. Furthermore, in addition to automatic evaluation and manual evaluation, we study in detail the model's concerns during the generation process through visualization.

Table 1. Complexity of implementations of two encoding methods

Strategy	Space complexity	Time complexity
Naive encoding	$O((l_1^k + l_2^k + \cdots + l_{num}^k)^2)$	$O((l_1^k + l_2^k + \cdots + l_{num}^k)^2)$
Bucket encoding	$O((N_1^b L^1)^2 + \cdots + (N_5^b L^5)^2)$	$O((N_1^b L^1)^2 + \cdots + (N_5^b L^5)^2)$

3 Method

We will introduce RoKGDS in this section. Figure 2 shows an overview of our model. RoKGDS consists of two modules: a bucket encoder (Sect. 3.1) and a hybrid decoder (Sect. 3.2). We initialize our model from RoBERTa. For each conversation turn, the bucket encoder obtains the feature extracted from the external knowledge, such as user profile, knowledge graph, and dialog goal. Then the hybrid decoder incorporates the context and knowledge features to generate the final dialog system response.

3.1 Bucket Encoder

In the knowledge grounded dialog task, the knowledge involved is always massive and diverse. To efficiently encode knowledge, we use a *bucket encoding* method. We first concatenate structured knowledge into the form of natural language (e.g. knowledge triple (h, r, t) to $[h; r; t]$). Then we put knowledge texts of similar length into the same bucket. We set five buckets with maximum lengths L of 16, 32, 64, 128, and 256, respectively. We use a Transformer to process knowledge in the same bucket in parallel. Specifically, the process is as follows:

$$K_i = \{k_j \big| L_{i-1} < |k_j| \leq L_i\} \tag{1}$$

$$B_i = \text{Transformer}(K_i) \tag{2}$$

where k_j is j-th knowledge text. Finally, the bucket encoder outputs E_K (the concatenation of all the encoded knowledge feature $e_{z,i}^k$ contained in B_i) for next stage processing. Given the knowledge texts $[k_1, k_2, ..., k_{num}]$ with the lengths $[l_1^k, l_2^k, ..., l_{num}^k]$ and the number of knowledge texts contained in each bucket after splitting operation $[N_1^b, N_2^b, ..., N_5^b]$. The complexities of naive encoding and bucket encoding are summarized in Table 1. Table 1 shows that bucket encoder is more efficient to encoding knowledge than naive encoding.

3.2 Hybrid Decoder

To enhance the robustness of the model, we develop a hybrid attention in the hybrid decoder to adapt the Chinese pre-trained language model RoBERTa to a knowledge grounded dialog system. Moreover, we use a copy mechanism [13] to copy the given knowledge. The hybrid attention and copy mechanism help to generate appropriate responses with correct knowledge for scenarios that never

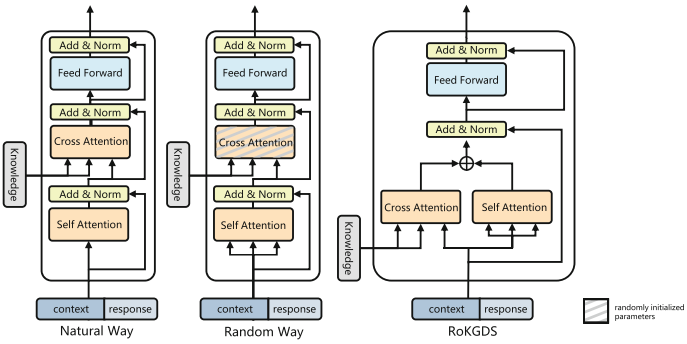

Fig. 3. Adaptation schemes

appeared during the training process. The input to the hybrid decoder is the knowledge feature E_K and the concatenation of a context and response $s = (x_1, ..., x_N, [sep], y_1, ..., y_M)$.

Adaptation Schemes. Considering that there is a gap between the RoBERTa pre-training task and the knowledge grounded dialog task, we propose the hybrid attention to combine our model with the pre-trained language model. As shown in the right of Fig. 3, the hybrid attention processes self attention and cross attention in the same layer, and the averaged output is input in the feed-forward network. Additionally, we study the other two adaptation schemes for comparison. First, a natural way to adapt RoBERTa to the Transformer based encoder-decoder framework is successively processing context and knowledge information by the multi-head attention with the pre-trained parameters. Although this natural way seems effective, it is difficult to fit the training data. So we study the second way (called random way) that we randomly initialize the parameters of the cross-attention. The random way performs better than the natural way shown in experiments.

Copy Mechanism. Due to the small amount of training data, the model tends to overfit the training data, leading to the model not be able to make good use of the knowledge it has never seen before. Although we input knowledge related to the current scenario, the model will produce responses with unrelated knowledge. To address this problem, we use the copy mechanism to copy the input knowledge appropriately. Specifically, we compute the word generation

Table 2. Dataset statistics of the three tasks

Task	Dataset	Knowledge graph	Profile	Goal	Train	Dev	Test
Knowledge dialog	DuConv	✓	✗	✓	19,858	2,000	5,000
Recommendation dialog	DuRecDial	✓	✓	✓	6,618	946	4,645
Persona dialog	CPC	✗	✓	✗	23,000	1,500	1,500

probability $P(w)$ at step t as follows:

$$score = \text{Softmax}(E_K O_{dec}^t) \tag{3}$$

$$P_g = \text{Softmax}(W_{voc} O_{dec}^t) \tag{4}$$

$$\alpha = \text{Sigmoid}(W_k O_{dec}^t) \tag{5}$$

$$P(w) = \alpha * P_g(w) + (1 - \alpha) \sum_{i:w=w_i} score_i \tag{6}$$

where W_{voc} denotes the word embedding matrix and W_k is a trainable parameter. O_{dec}^t is the outputs of the Transformer in hybrid decoder at step t. $\alpha \in [0, 1]$ controls the weight of generating a general word.

Training Objective. We optimize the following cross-entropy loss between the predicted word distribution P and ground-truth distribution o:

$$L = -\sum_{j=1}^{M} o_j \log\left(P(y_j)\right) \tag{7}$$

4 Experiments

4.1 Dataset

2021 Language and Intelligence Challenge: Multi-skills Dialog Task contains the tasks of the three major aspects of the dialog: knowledge dialog [17], recommendation dialog [10], and persona dialog. The knowledge dialog task requires the model to generate fluent, appropriate responses containing correct knowledge. The recommendation dialog task requires the model to appropriately respond to user requests, and proactively make recommendations based on user interests. The persona dialog task requires the model to generate fluent, appropriate, and persona-consistent responses. We show the details of the datasets used in the above three tasks in Table 2.

4.2 Implement Details

We implement RoKGDS in PyTorch[1]. Both the bucket encoder and hybrid decoder contain 12 Transformer blocks. Each Transformer block uses 12 attention

[1] Code will be available at https://github.com/z562/RoKGDS.

Table 3. Training details

Task	Learning rate	Batch size	Epoch	GPU	Training time
Knowledge dialog	4.0e−5	4	8	16 G Tesla P100 GPU	33 h
Recommendation dialog	3.5e−5	4	6		30 h
Persona dialog	2.5e−5	12	8		80 h

Table 4. Automatic evaluation

Method	DuConv	DuRecDial	Persona
	F1/BLEU-1/BLEU-2	F1/BLEU-1/BLEU-2	F1/BLEU-1/BLEU-2
Transformer	36.99%/0.302/0.176	56.40%/0.429/0.358	27.58%/0.270/0.145
GPT-2	43.87%/0.344/0.202	60.02%/0.494/0.419	34.11%/0.319/0.185
RoKGDS	**46.74%/0.376/0.252**	**63.43%/0.545/0.468**	**35.00%/0.329/0.201**
RoKGDS+Nat	0.57%/0.000/0.000	1.43%/0.000/0.000	1.91%/0.000/0.000
RoKGDS+Ran	44.59%/0.350/0.223	61.13%/0.513/0.441	33.62%/0.313/0.186
RoKGDS-Copy	46.04%/0.372/0.249	62.37%/0.527/0.451	34.64%/0.320/0.192

heads. The word embedding size is set to 768. The vocabulary size is 21,128. The maximum context length is 512. We use Adam [7] algorithm with the mini-batch method for optimization. We show the settings on different tasks in Table 3.

4.3 Baseline Models

We compare RoKGDS against several baselines:

- **Transformer:** We implement the Transformer model as introduced by Radford et al. [15].
- **GPT-2:** We initialize RoKGDS without the copy mechanism from the GPT-2 [12].

To further study the impact of various adaptation schemes and components on the model, we conduct the following ablation experiments: (1) **RoKGDS+Nat**: with the adaptation scheme *natural way*; (2) **RoKGDS+Ran:** with the adaptation scheme *random way*; (3) **RoKGDS-Copy**: without the copy mechanism.

4.4 Automatic Evaluation

Metrics. We use F1 [10] and BLEU [11] to evaluate the models. F1 measures the token-level similarity between the predicted response and the ground truth, while BLEU evaluates the word-level similarity.

Results. We present the results of automatic evaluation in Table 4. Our model achieves a significant improvement over other baseline models. This indicates that with the bucket encoder and the hybrid decoder, our proposed model effectively extracts the knowledge and incorporates the knowledge into the answer generation, lead to high robustness on the test datasets. We can also find that

Table 5. Human evaluation

Method	DuConv			DuRecDial			Persona		
	Info.	Cohe.	Know.	Info.	Cohe	Know.	Info.	Cohe.	Pers.
Transformer	1.325	1.105	1.065	1.570	1.510	1.360	1.190	1.025	1.005
GPT-2	1.590	1.545	1.460	1.625	1.590	1.685	1.505	1.430	1.105
RoKGDS	**1.660**	**1.736**	**1.520**	**1.845**	**1.895**	**1.740**	**1.550**	**1.580**	**1.315**

the score of the model with randomly initialized parameters has a big gap compared to the other models, indicating the pre-trained language model (GPT-2 or RoBERTa) can improve the model's performance in all kinds of knowledge grounded tasks. We show the results of the ablation study in 4. Notably, the performance of RoKGDS+Nat has extremely poor results. After our careful inspection, the RoKGDS+Nat is difficult to fit the training data. This is because the model cannot extract context information effectively. We use visualization to conduct detailed analysis in Sect. 4.6. We also find without the copy mechanism, the performance of our model has declined on all tasks, especially on the tasks that require rich knowledge to generate responses, indicating that the copy mechanism improves the utilization of input knowledge.

4.5 Human Evaluation

Metrics. We conduct human evaluations to compare three models, Transformer, GPT-2, and RoKGDS. We randomly sample 200 examples from the test set of the three tasks. Given the context and related knowledge, we generate the responses use all three models. Then we employ three human evaluators to assess the responses. For knowledge dialog and recommendation dialog, the human evaluators measure the responses in terms of informativeness, coherence, knowledge consistency using a 3-point Likert scale. For persona dialog, the human evaluators measure the responses in terms of informativeness, coherence, persona consistency. The informativeness measures if the generated responses are contentful. The coherence measures if the generated responses are appropriateness and coherence of the whole dialog flow. The knowledge consistency measures if the knowledge used in responses is correct and proper. The persona consistency measures if the responses match the given profile.

Results. The results of human evaluation are shown in Table 5. Our model outperforms all the baselines, indicating that our model can generate more appropriate and informative responses. We also observe that our model has a better performance in knowledge-rich dialog tasks (knowledge dialog and recommendation dialog). This is because our model can build a good connection between knowledge and context so that it has better performance in test scenarios.

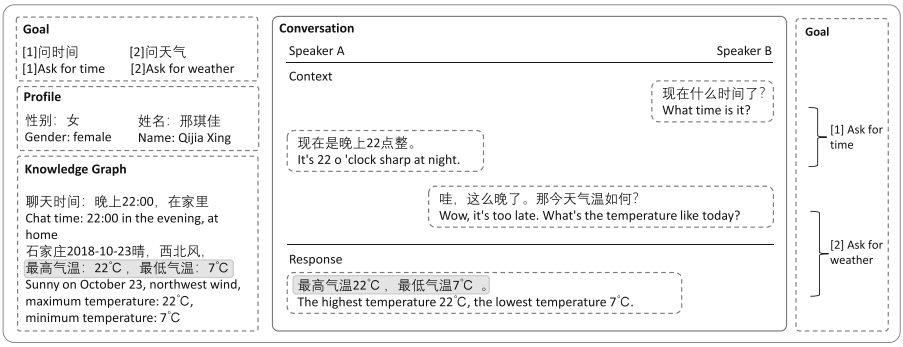

Fig. 4. A recommendation dialog sample for visualization.

4.6 Visualization

To show the behavior of the three adaptation schemes in response generation, we provide a recommendation dialog sample. As shown in Fig. 4, the model needs to use the knowledge graph information to generate the correct response *"the maximum temperature 22℃, the minimum temperature 7℃"*. We visualize the maximum response-context attention score in Fig. 5 and the maximum response-knowledge attention score in Fig. 6. As shown in the left group of Fig. 5, RoKGDS+Nat has very low maximum attention scores for the context in the second to seventh layers of Transformer compared to the first layer, indicating that the model cannot effectively use context information in these layers. This is because in the pre-training stage the RoBERTa only contains one multi-head attention layer, while in the fine-tuning stage our model uses two multi-head attention layers to process context and knowledge in turn. This leads to the continuous dilution of the dialog history information after the knowledge information is input in the first layer. In the early stage of model training, it will have a destructive effect on model fitting, resulting in very poor performance. The middle groups in Fig. 5 shows that RoKGDS+Ran can extract contextual information normally. This is because randomly initialized cross-attention weights alleviate the influence of the input of knowledge information and reduce the gap between the pre-training stage and fine-tuning stage. However, RoKGDS+Ran pays less attention to the knowledge information and tends to use the dialog context information to generate responses, which reduces the robustness of the model. It can also be reflected in the results of automatic and human evaluations. Notably, RoKGDS focuses on the dialog context and the input knowledge information simultaneously, which suggests that the model has higher robustness than the other models, and can use the knowledge that the model has never been seen before to generate correct and appropriate responses based on the context.

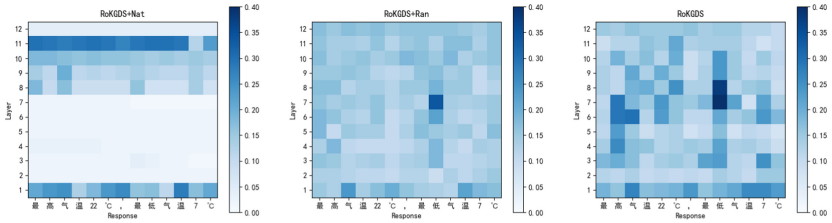

Fig. 5. Visualization of the maximum attention score for each token in the response to focus on the dialog context

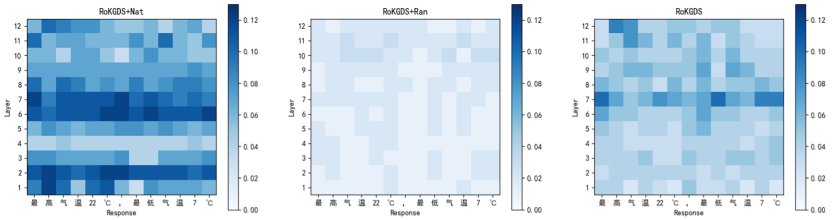

Fig. 6. Visualization of the maximum attention score for each token in the response to focus on the related knowledge

4.7 Conclusions

In this work, we propose the RoKGDS to enhance the model's robustness in knowledge grounded dialog task. We use a bucket encoder to efficiently extract knowledge features, such as profile, knowledge graph, and dialog goal. Then we propose a hybrid decoder with a hybrid attention and a copy mechanism to generate informative and proper responses in the unseen scenarios. Both automatic and human evaluation verifies the effectiveness of the proposed model. To further study the three adaptation schemes, we use the visualization and explain how the model performs in automatic evaluation.

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